

A sample of an unknown ionic salt with a mass of m_{salt} is added to a calorimeter containing a sample of water with a mass of $m_{\text{H}_2\text{O}}$. Which expression gives the total amount of heat absorbed by the solution during the dissolution reaction if the initial temperature of the solution is T_{initial} and the final temperature is twice that of T_{initial} ? (Note: The specific heat capacity of the solution is C_s .)

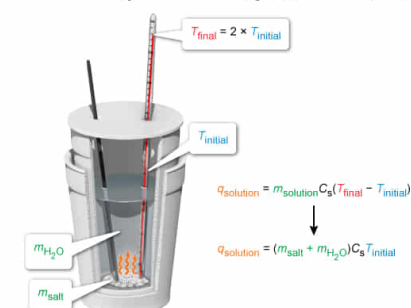
- ✓ ☒ A. $(m_{\text{salt}} + m_{\text{H}_2\text{O}})C_s T_{\text{initial}}$ [60%]
☐ B. $m_{\text{salt}}C_s(T_{\text{initial}})$ [14%]
☐ C. $(m_{\text{salt}} + m_{\text{H}_2\text{O}})C_s(2T_{\text{initial}})$ [20%]
☐ D. $m_{\text{H}_2\text{O}}C_s(-T_{\text{initial}})$ [4%]

Correct

60% answered correctly

Explanation:

Heat absorbed by a solution during a dissolution reaction



During a **dissolution** reaction, the amount of heat q released (indicated by a negative sign) or absorbed (indicated by a positive sign) by the reaction is **equal in magnitude** but **opposite in sign** to the amount of heat absorbed or lost by the solution.

The question states that an unknown salt (ie, the solute) with a mass of m_{salt} is dissolved in a sample of water (ie, the solvent) with a mass of $m_{\text{H}_2\text{O}}$, causing the temperature of the solution to increase from T_{initial} to twice that of T_{initial} (ie, $2 \times T_{\text{initial}}$). This increase indicates that the reaction releases heat, and the solution absorbs the heat:

$$-q_{\text{reaction}} = q_{\text{solution}}$$

The amount of heat absorbed by the solution resulting in the temperature change ΔT depends on its mass ($m_{\text{solution}} = m_{\text{solute}} + m_{\text{solvent}}$), **specific heat capacity** C_s , final temperature T_{final} , and initial temperature T_{initial} according to the relationship:

$$q_{\text{solution}} = m_{\text{solution}}C_s \Delta T$$

$$q_{\text{solution}} = (m_{\text{solute}} + m_{\text{solvent}})C_s(T_{\text{final}} - T_{\text{initial}})$$

Therefore, substituting the appropriate masses and $T_{\text{final}} = 2 \times T_{\text{initial}}$ into the above equation yields the expression for the total amount of heat absorbed by the solution:

$$q_{\text{solution}} = (m_{\text{salt}} + m_{\text{H}_2\text{O}})C_s(2T_{\text{initial}} - T_{\text{initial}})$$

$$q_{\text{solution}} = (m_{\text{salt}} + m_{\text{H}_2\text{O}})C_s(T_{\text{initial}})$$

(Choices B and D) The mass of the solution is equal to the mass of the solute *plus* the mass of the solvent. In addition, if ΔT is expressed as $T_{\text{initial}} - T_{\text{final}}$ rather than $T_{\text{final}} - T_{\text{initial}}$, ΔT incorrectly equals $-T_{\text{initial}}$ (ie, the solution loses heat).

(Choice C) The expression $(m_{\text{salt}} + m_{\text{H}_2\text{O}})C_s(2T_{\text{initial}})$ is obtained if it is incorrectly assumed that the heat absorbed by the solution depends only on the final temperature (ie, $T_{\text{final}} = 2 \times T_{\text{initial}}$) rather than the **change** in temperature.

Educational objective:

The heat q absorbed by a solution during a dissolution reaction depends on the solution's mass m_{solution} , its specific heat capacity C_s , and the change in temperature ΔT , according to the equation $q_{\text{solution}} = m_{\text{solution}}C_s\Delta T$.

Time spent:0 sec
Subject:
General Chemistry

QID:403896
System:
Thermodynamics, Kinetics & Gas Laws